

# KALKULUS SECURYTAS

Universal Temporary Cross-Device Data & File Storage System

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## 1. System Overview & Purpose

Kalkulus Securytas is a minimalist, ultra-fast cross-device clipboard and file sharing application. It enables users to transfer text snippets, full source code repositories, PowerPoint presentations, videos, audios, PDFs, and documents between any device or browser using a simple 4-digit numeric PIN code—without requiring account creation or login.

Key Requirement: 100% Temporary Storage. Every uploaded item has a configurable expiration timer (1 Hour, 24 Hours, 3 Days, 7 Days). Once expired, the background server cleanup routine automatically unlinks files from the disk and removes data from memory.

## 2. Technology Stack & Architecture

- **Frontend Layer:** Built using HTML5, Vanilla ES6+ JavaScript, and modern CSS3 (Pitch-Black Minimalist design system, CSS Flexbox & Grid, responsive mobile media queries). Uses FontAwesome font icons and QRCode.js for mobile camera scanning.
- **Backend API Server:** Powered by Node.js and Express.js REST APIs handling CORS, JSON body payloads (up to 100MB body parser limits), file downloads, and PIN queries.
- **File Upload Engine:** Utilizes Multer middleware with disk storage engines for handling multipart file uploads (supporting PPTX, Video, Audio, PDF, Docs, Zip).
- **Data Persistence & Cleanup:** JSON file-backed data store (``data_store.json``) with automatic fallback to ``/tmp`` directory on Vercel serverless environments. Background interval process runs every 60 seconds to purge expired clips.
- **Deployment Compatibility:** Pre-configured with ``vercel.json`` for 1-click Vercel Serverless deployment, as well as 24/7 persistent hosting on Render, Railway, Koyeb, or local Node.js environment.

## 3. Step-by-Step Workflow

**Step 1: Data / File Upload (Sender Device):** Sender pastes text/code or drops a file (PPT, Video, Audio, PDF) and selects an expiration timer (1 Hour, 24 Hours, 3 Days, 7 Days).

**Step 2: PIN Code & WhatsApp Generation:** The server generates a unique 4-digit numeric PIN (e.g. 4829) and formats a 1-click WhatsApp share link with the PIN and direct URL.

**Step 3: PIN Retrieval (Receiver Device):** On any receiving device or browser, the user enters the 4-digit PIN code. The server matches the PIN, increments view metrics, and opens the payload with 1-click Copy or File Download options.

**Step 4: Automatic Disk Purge:** When the clip reaches its expiration timestamp, the server unlinks the file from disk (``fs.unlinkSync``) and purges records from data storage.

## 4. REST API Technical Reference

**POST /api/clip** - Creates text, code, or URL clip. Returns { pin, id, clip }.

**POST /api/upload** - Uploads file payload via Multer multipart form. Returns { pin, id, clip }.

**GET /api/pin/:pin** - Fetches active clip payload by 4-digit numeric PIN.

**GET /api/clips** - Returns list of active non-expired storage vault items.

**DELETE /api/clips/:id** - Manually deletes a clip and unlinks file from server storage.

**GET /api/download/:id** - Streams file download attachment with original filename.

**GET /api/stats** - Returns server metrics (total clips, total storage used in MB, uptime).